Ankur Karn

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Education

UPES (University of Petroleum and Energy Studies) Dehradun, India
B.Tech, Computer Science & Engineering – Cyber Security & Forensics *Aug 2023 – May 2027*

Experience

Pheme Software Pvt. Ltd. (IBM Platinum Business Partner) Remote
Security Intern – Mobile Application Security Analysis System *Jun 2026 – Aug 2026*
- Built a **Mobile Application Security Analysis System** to assess mobile apps for security weaknesses.

Karnstack (karngyan.com) Remote
AI Automation Intern *Aug 2026 – Sep 2026*
- Built an end-to-end **AI content-creation workflow** using **prompt engineering**, chaining Claude (LLM) steps from topic to publish-ready draft.
- Created reusable prompt templates that kept tone and quality consistent and cut manual writing effort.

ZeroError Education Remote
Web Developer & SEO Intern *Jun 2024 – Jul 2024*
- Built WordPress/HTML/CSS pages and applied on-page SEO to improve search ranking.

Projects

CampusGPT – AI Campus Helpdesk Chatbot | *Python, FastAPI, PostgreSQL, pgvector, Next.js, LLMs, RAG*
- Built a full-stack AI assistant that answers students' questions on attendance, exams, fees, timetables and university rules in **English and Hinglish**, streaming answers in real time.
- Designed a **secure tool-calling architecture**: the LLM never touches the database and only calls read-only tools bound to the logged-in student, so it cannot read another student's data.
- Implemented **hybrid RAG** over policy documents (BGE embeddings with a pgvector HNSW index, plus PostgreSQL full-text search, fused by Reciprocal Rank Fusion) with **citation verification**; the right section is in the top 4 results for **96%** of benchmark questions, in under 11 ms.
- Ran the LLM **fully locally** (Qwen3 4B via Ollama on a laptop GPU), switchable to a hosted model (GPT-OSS 120B on Groq) through a provider adapter.
- Hardened against **prompt injection** and data leaks: an automated red-team suite (impersonation, jailbreaks, smuggled tool arguments, SQL injection) blocks **9/9 attacks**; added role-based access, audit logs and login/chat rate limiting.
- Built a synthetic university of **5,000 students**, an admin document pipeline with a background worker, a student dashboard, **250+ automated tests** and a 50-question benchmark against plain-LLM, RAG-only and database-only variants.

Cognitive Honeypot with AI Integration | *Python, Isolation Forest*
- Built a honeypot that simulates vulnerable services to lure attackers, with an Isolation Forest model that flags anomalous attack patterns in real time.

SQL Injection Detection using AI | *Python, Machine Learning*
- Designed an AI-based detector that analyses query patterns to flag SQL injection attempts.

DDoS Attack Detection using Traffic Pattern Analysis | *Python, Networking*
- Developed a network traffic analysis tool that detects DDoS attacks from abnormal traffic spikes and behaviour.

Smart Hospital & Inventory Management Systems | *Java, MySQL, JDBC*
- Built a hospital system (patient registration, appointments, staff management) and a desktop inventory system with real-time stock tracking and restocking alerts.

Technical Skills

Languages: Python, Java, C++, SQL, TypeScript, HTML, CSS

AI / ML: LLM applications, RAG, tool calling, embeddings & vector search, prompt engineering, prompt-injection defence, Claude, Ollama, scikit-learn

Web & Backend: FastAPI, Next.js, React, Tailwind CSS, REST APIs, Server-Sent Events, WordPress

Databases: PostgreSQL (pgvector, full-text search), MySQL, MongoDB

Security: Burp Suite, SQLMap, Metasploit, OWASP ZAP, Linux, threat modelling, secure design

Tools: Git, GitHub, Docker, pytest, VS Code

Achievements & Certifications

- Internship certificate, Mobile Application Security Analysis System – Phemesoft (IBM Platinum Business Partner), 2026.
- Solved 100+ problems on LeetCode (data structures and algorithms).